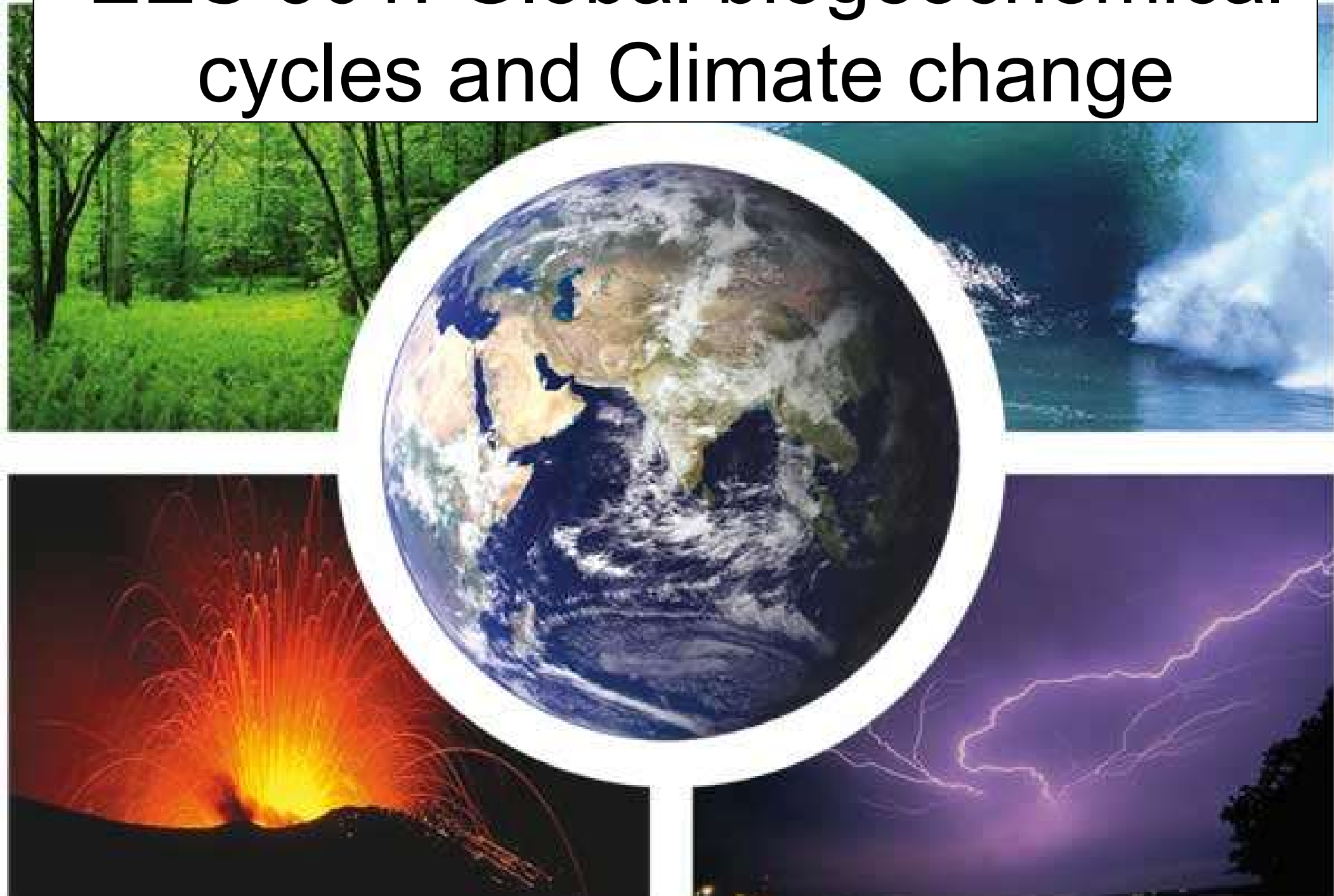


EES 301: Global biogeochemical cycles and Climate change



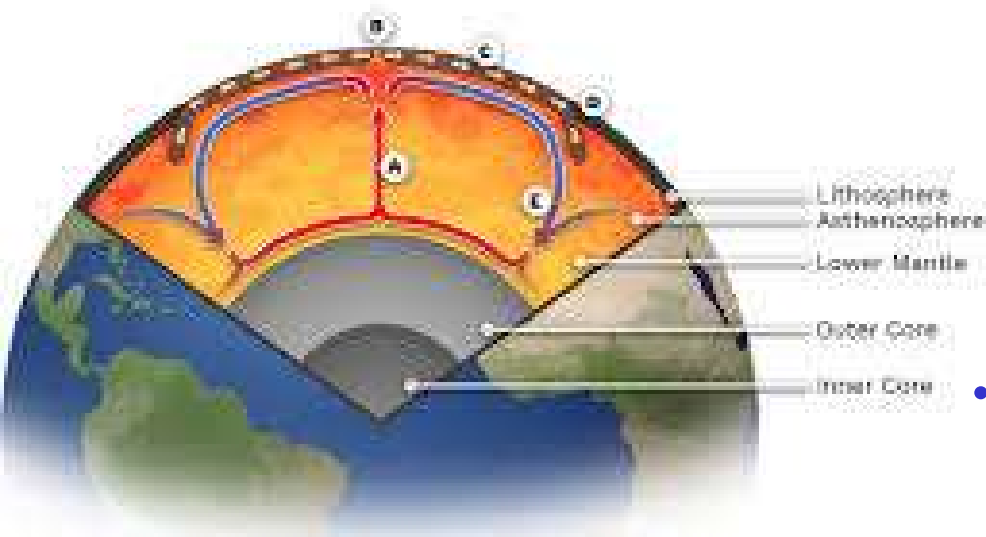
The Earth system is a closed system with respect to atomic matter and governed by the laws of mass conservation



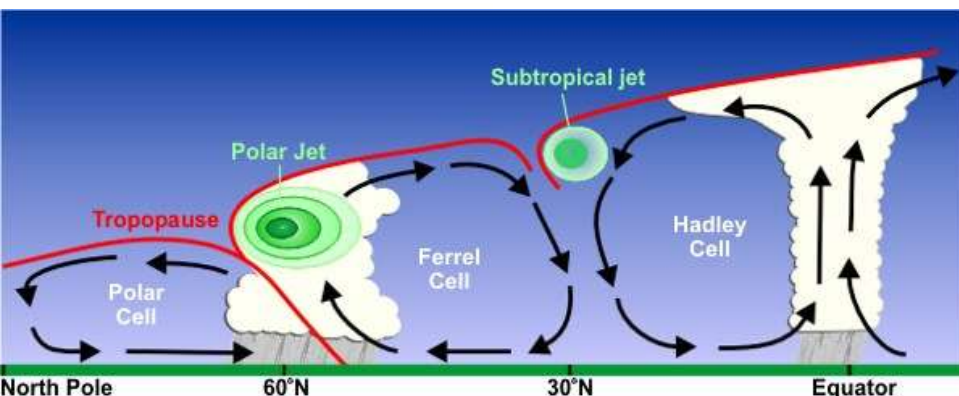
- Virtually all of the atoms in the solid, liquid and gaseous parts of the Earth have been part of the planet for all its approximately 4.5 billion years of existence
- Very few have changed (by radioactive decay), departed to space or arrived via meteorites after the Hadean
- **Pollution is therefore a human-induced change in the distribution of atoms from one place of the Earth to another. This definition includes the manufacturing of new molecular entities.**

<https://earthobservatory.nasa.gov/images/885/earth-from-space>

Energy from the sun and from radioactive decay in the Earth's core drive cycling of materials through different reservoirs in the Earth System



- Almost all rocks have been processed at least once through a chemical and physical cycle involving weathering, formation of sediments and subduction, being subjected to great heat and pressure to form new igneous or metamorphic rocks (**Rock cycle**)
- The water in the oceans has been evaporated, rained out and returned via rivers and groundwater flow many tens of thousands of times (**Hydrological cycle**)
- The Main gases in the atmosphere (nitrogen and oxygen) are cycled through living organisms frequently
- **Anthropogenic changes are superimposed on the natural cycles**



The biogeochemical cycles describe the movement of specific elements (C, N, O, S..) through the different geospheres on our planet



- The **biosphere** is the collection of the biota (all living things) on the planet
- The **hydrosphere** includes all oceans, freshwater body and all other water reservoirs
- The **atmosphere** is the gas-phase envelope surrounding our globe the Earth
- The **lithosphere** is the entirety of rocks on Earth
- The **pedosphere** (literally that upon which we walk) comprises the soils of the Earth

The biogeochemical cycles describe the movement through the different geospheres on our planet

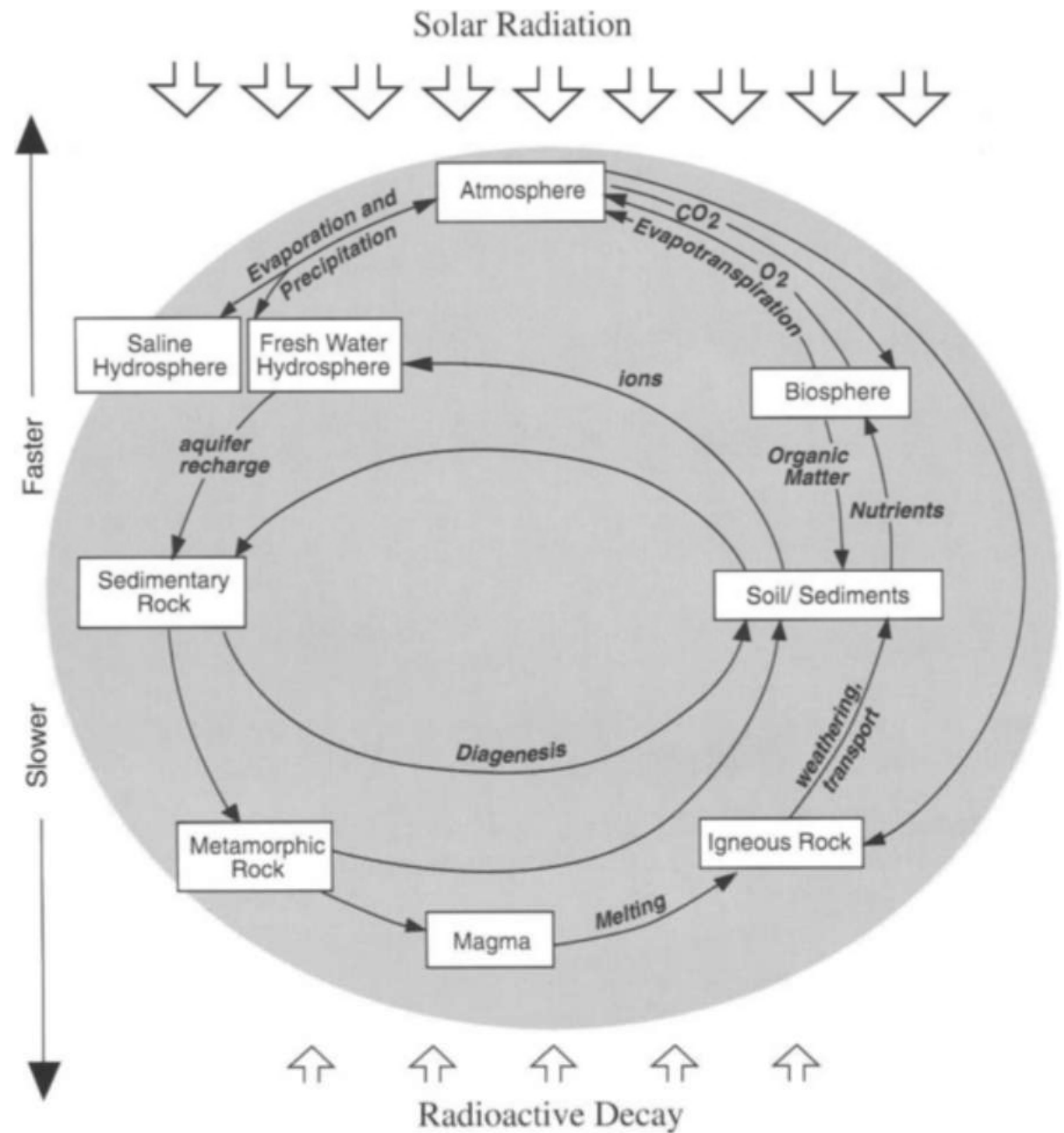


FIG. 1-1 Diagram of cyclic processes and fluxes between the major reservoirs on Earth.

Reservoirs, flux and burden

Reservoir (box, compartment):

An amount of material defined by certain physical, chemical or biological characteristics that for the purpose of analysis we consider to be reasonable homogeneous

Flux: The amount of material transferred from one reservoir to another per unit time, in general denoted by E with the units mass/time



Examples: Oxygen in the atmosphere, Carbon in living organic matter in the ocean, Sulfur in sedimentary rocks



Examples: The rate of evaporation of water from the ocean surface to the atmosphere. The rate of deposition of phosphorus on marine sediments. The rate of N_2O oxidation in the stratosphere.



If the reservoir is defined by its physical boundaries the content of the specific element is called its burden.

We will denote this with the symbol M

The units of M will usually be units of mass although sometimes it could also be moles

Sources and Sinks

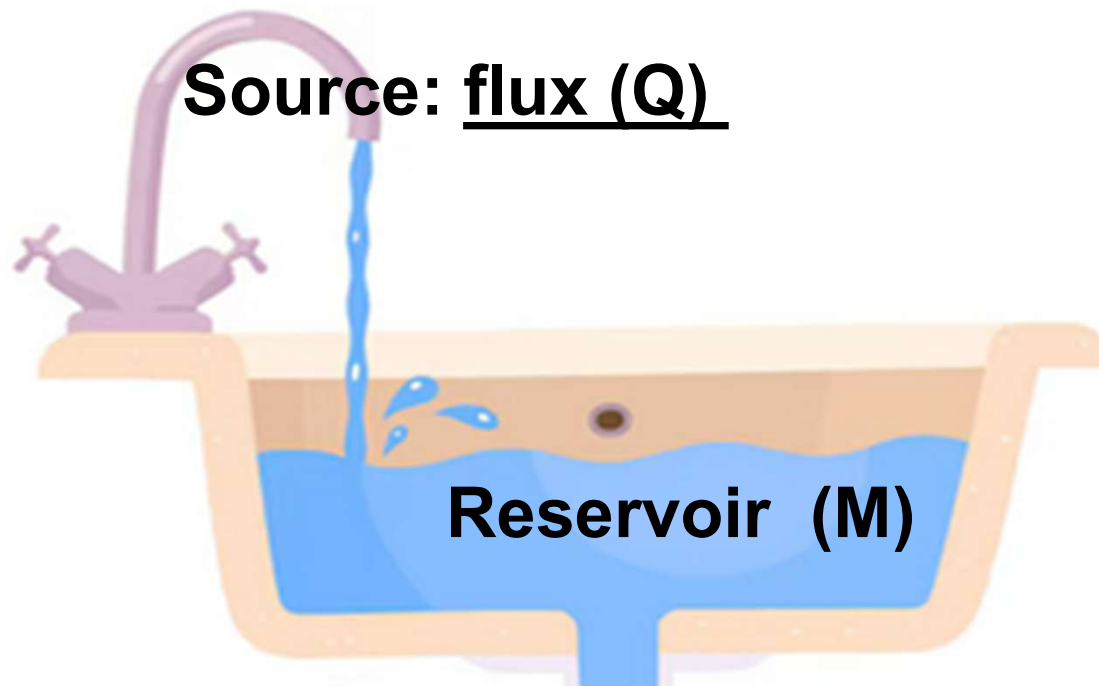
Source: A flux (Q) of material into a reservoir. Units are mass/time = g/s. The sign is positive.

Sink: A flux (S) of material out of a reservoir. Units are mass/time = g/s. The sign is negative.

Very often this flux is proportional to the burden of the reservoir $S=kM$. In such cases the sink flux is referred to as a first-order process.

If the sink flux is constant independent of the reservoir content the process is of zero order. Higher order fluxes, i.e. $S=kM^\alpha$ with $\alpha>1$ also occur.

$$\frac{dM}{dt} = Q - S$$



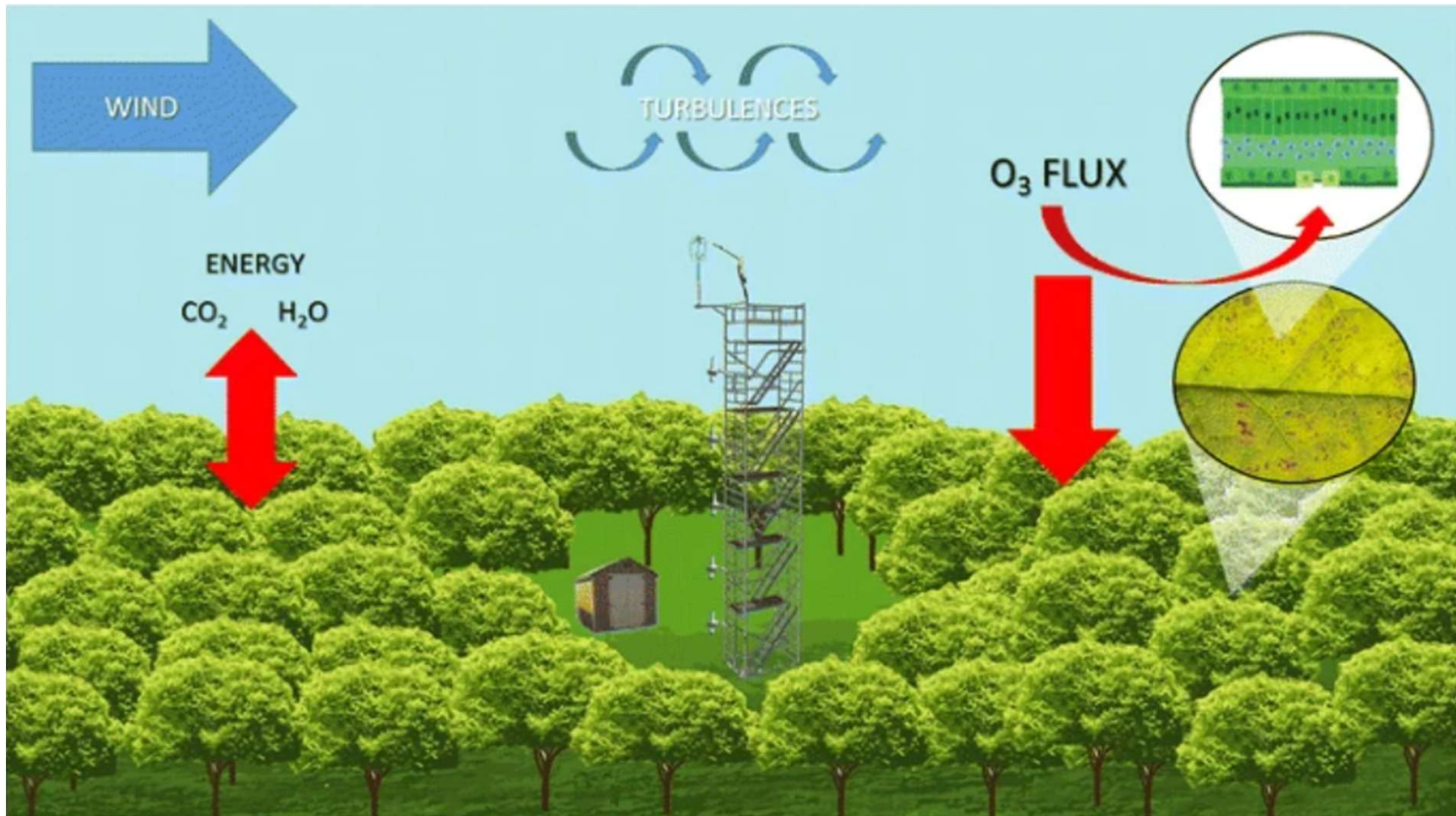
Source: flux (Q)

Sink: flux (S)

Flux density:

Flux density is used instead of flux in some transport processes (e.g. CO₂ assimilation through stomata): The amount of material transferred per unit area per unit time.

For stomatal exchange of gases the units are usually $\text{g m}^{-2} \text{s}^{-1}$ or $\text{mol m}^{-2} \text{s}^{-1}$ where the measured area is the photosynthetically active leaf surface area (dead leaves don't count).

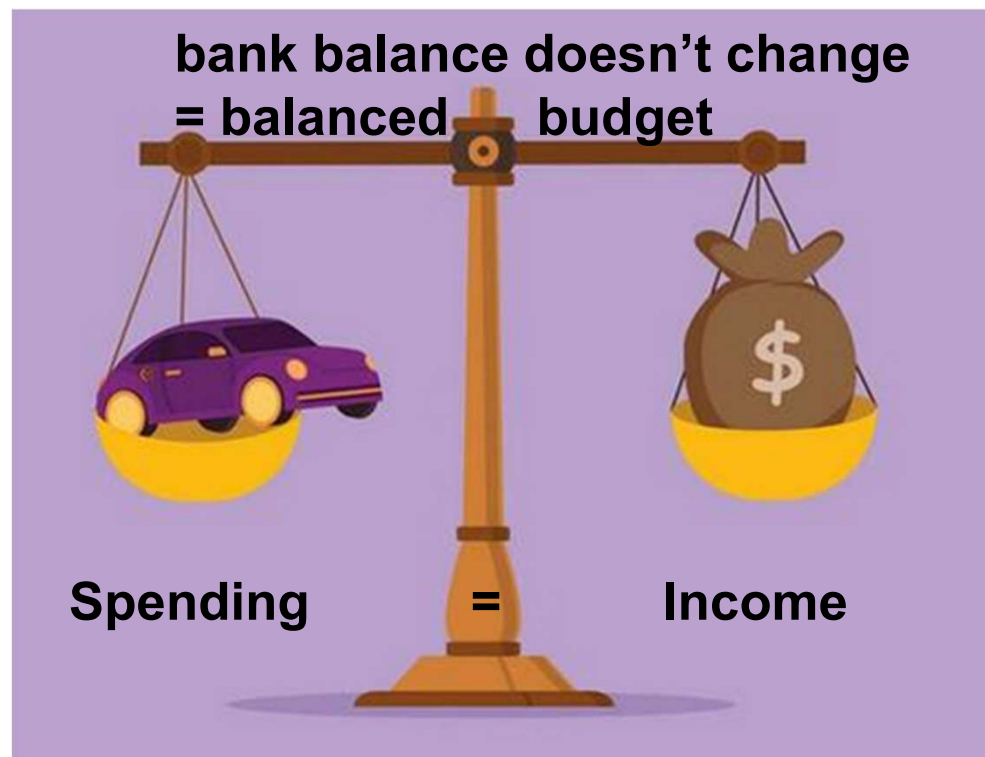


Budget:

A budget is balance sheet of all sources and sinks of a reservoir. If sources and sink balances and do not change with time the reservoir is in steady state, i.e., M does not change with time.

Usually some fluxes are better known than others. If steady state prevails, a flux that is unknown a priori can be estimated by difference from the other fluxes. If this is done it should be made clear during the presentation of the budget.

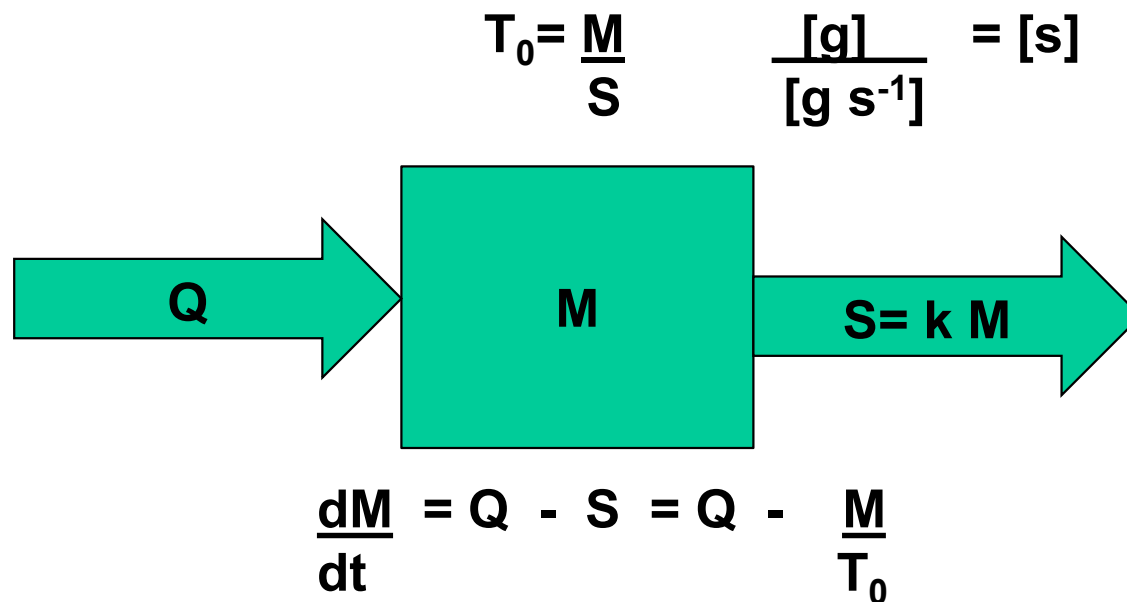
Budgets and cycles can be considered on very different spatial scales. We will consider global, hemispheric and regional scales. The choice of a suitable scale is determined by the homogeneity in the spatial distribution. E.g. the atmosphere can be considered one reservoir for O_2 or CO_2 but CO is too short lived and has different burden in the northern and southern hemisphere



Turnover time:

Turnover time or residence time: The turnover time τ_0 or residence time of a reservoir in steady state is the ratio of the content M of the reservoir to the sum of its sinks S or the ratio of M to the sources Q .

The turnover time is the time it would take to empty the reservoir in the absence of sources if the sinks remain constant and the average time a molecule stays in the reservoir.



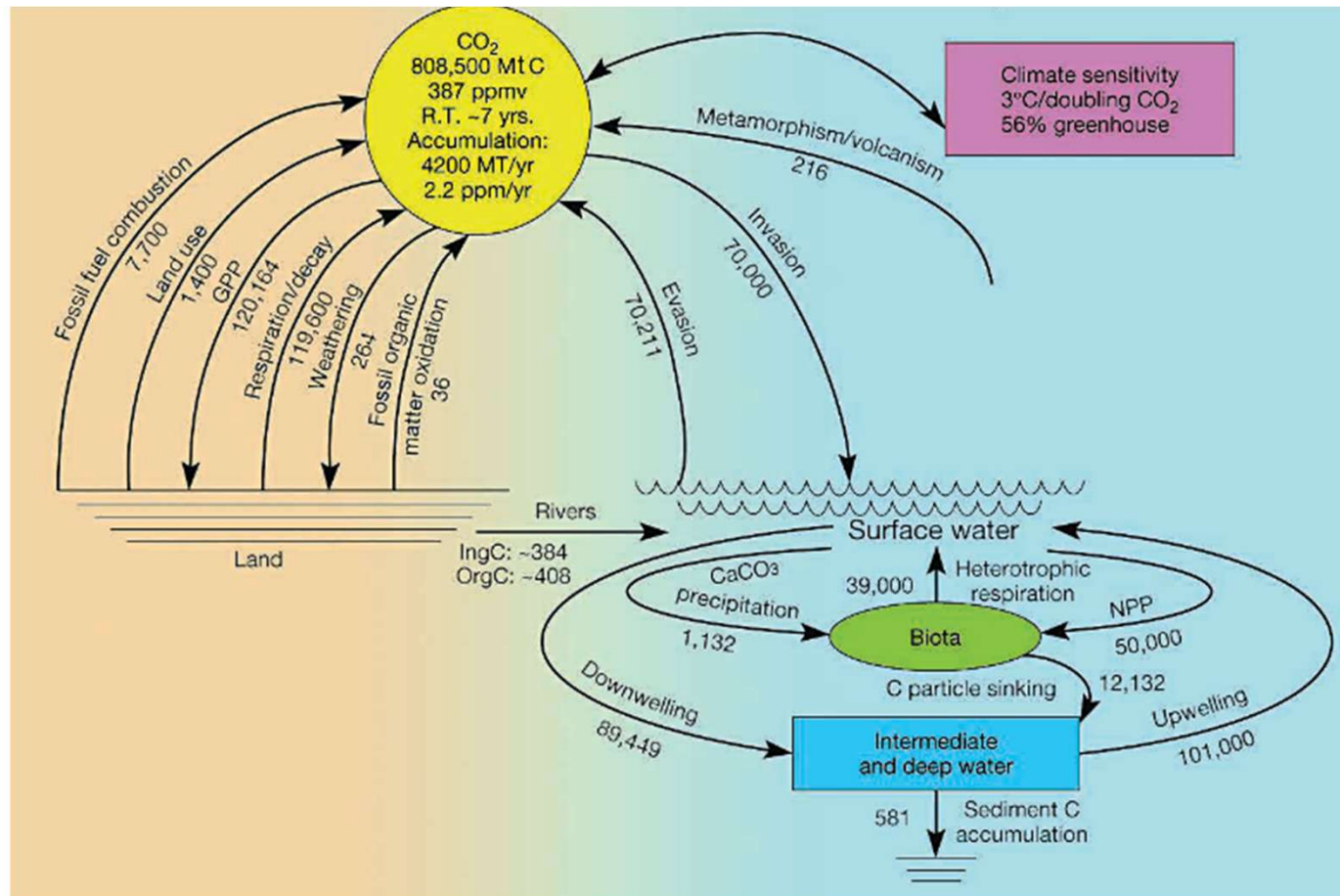
The turnover rate (fraction of the substance that leaves every year) is the inverse of the residence time (S/M) and often expressed in percent. Units $1/s$. If the residence time of students in the BS MS program is 5 years then on average $1/5$ or 20% of the students leave IISER every year.

Cycle:

Cycle: A system consisting of two or more connected reservoirs, where a large part of the material is transferred through the system in a cyclic fashion.

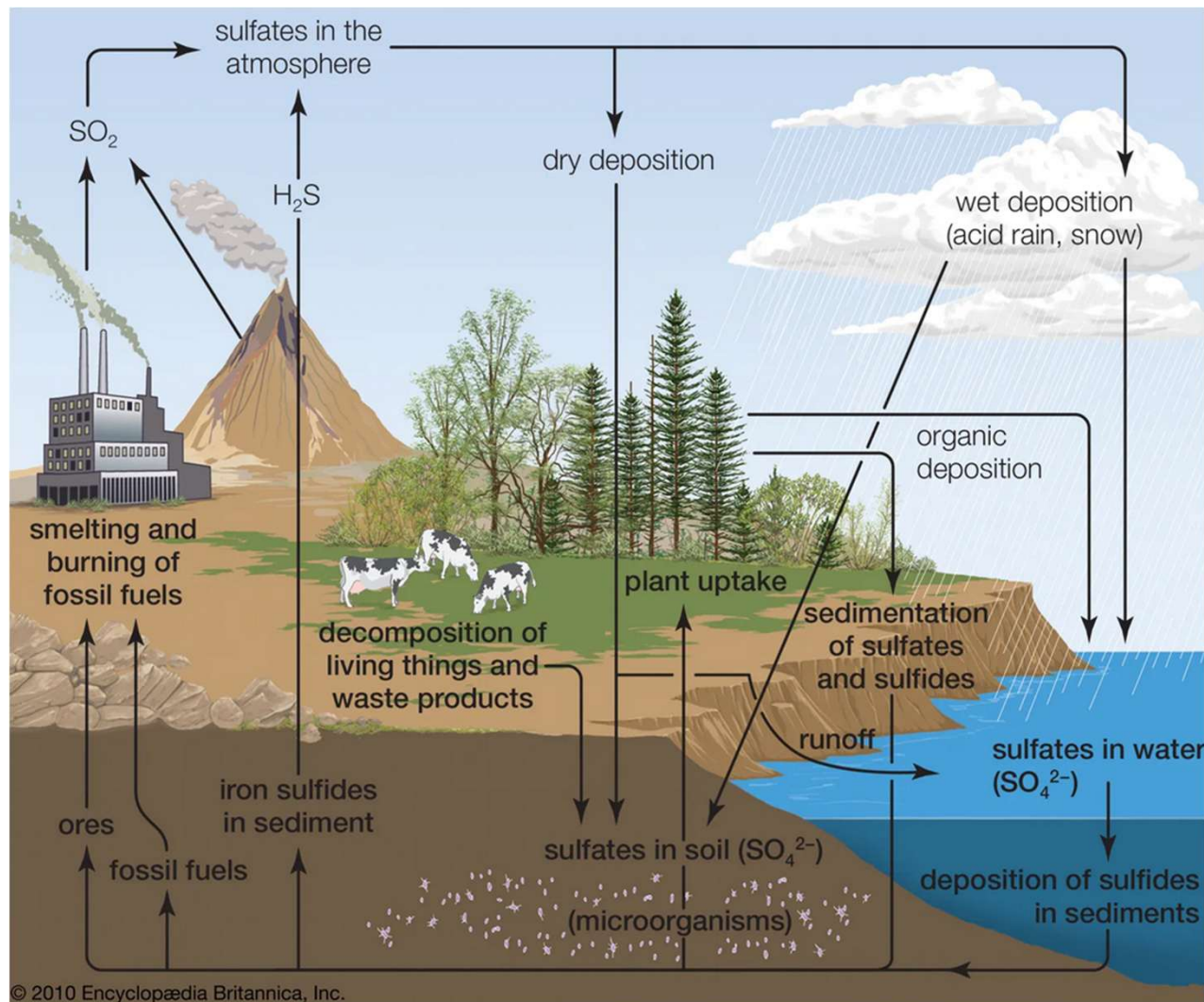
If all material cycles within the system then the system is closed.

Many systems of connected reservoirs are not cyclic but instead material flows unidirectional. In such systems some reservoirs (at the end of the chain) may be accumulative, whereas others remain balanced (non-accumulative).



Biogeochemical cycle

The term describes the global or regional cycles of elements such as C, N, S and P within reservoirs including the whole or part of the atmosphere, the ocean, the sediments and the living organisms.



Evidence for the coupled nature of the Earth System

Dead Planet:

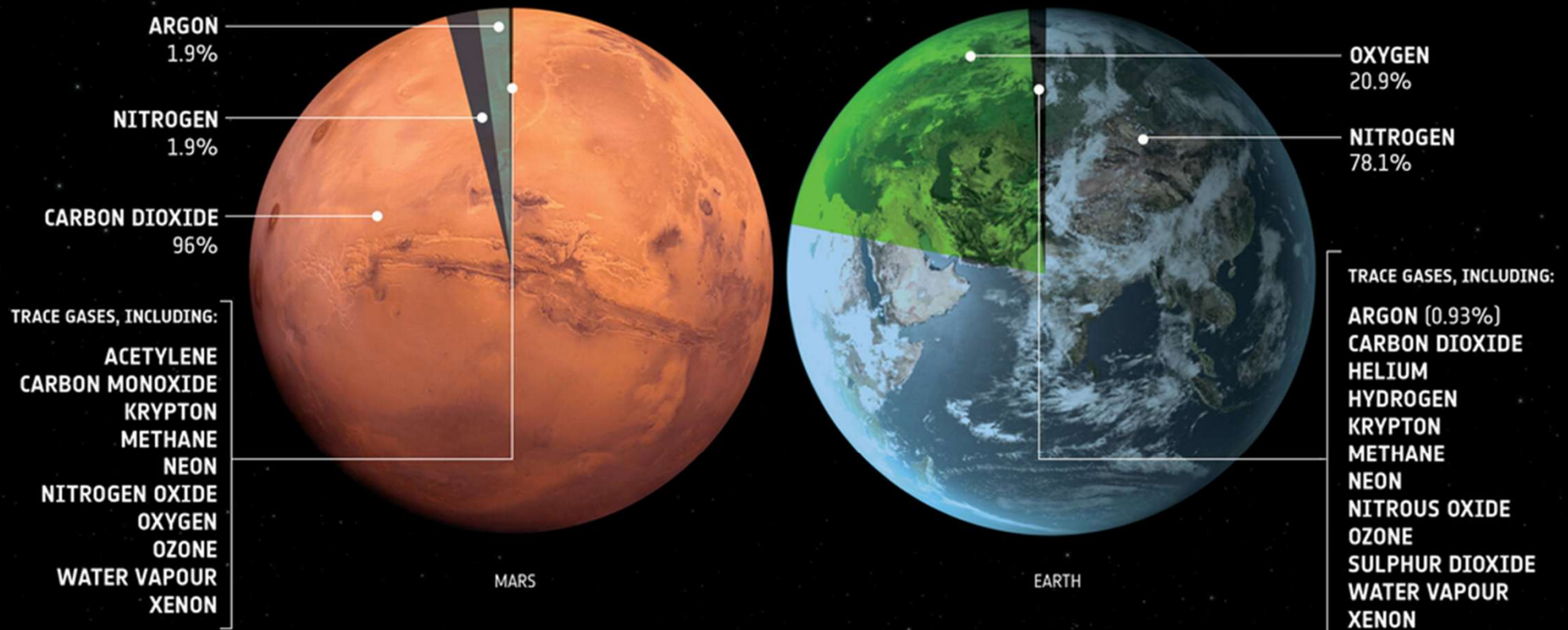
Near thermodynamic equilibrium
Calculated black body temperature
actual surface temperature 210 K
almost match

Planet full of life:

Out of thermodynamic equilibrium.
 O_2 , N_2 and H_2O coexist in the atmosphere equilibrium demands HNO_3 be formed and dissolved in the ocean.
 O_2 coexists with combustible biomass and reduced forms of iron, sulfur...



→ COMPARING THE ATMOSPHERES OF MARS AND EARTH



Evidence for the coupled nature of the Earth System

— warm age — ice age

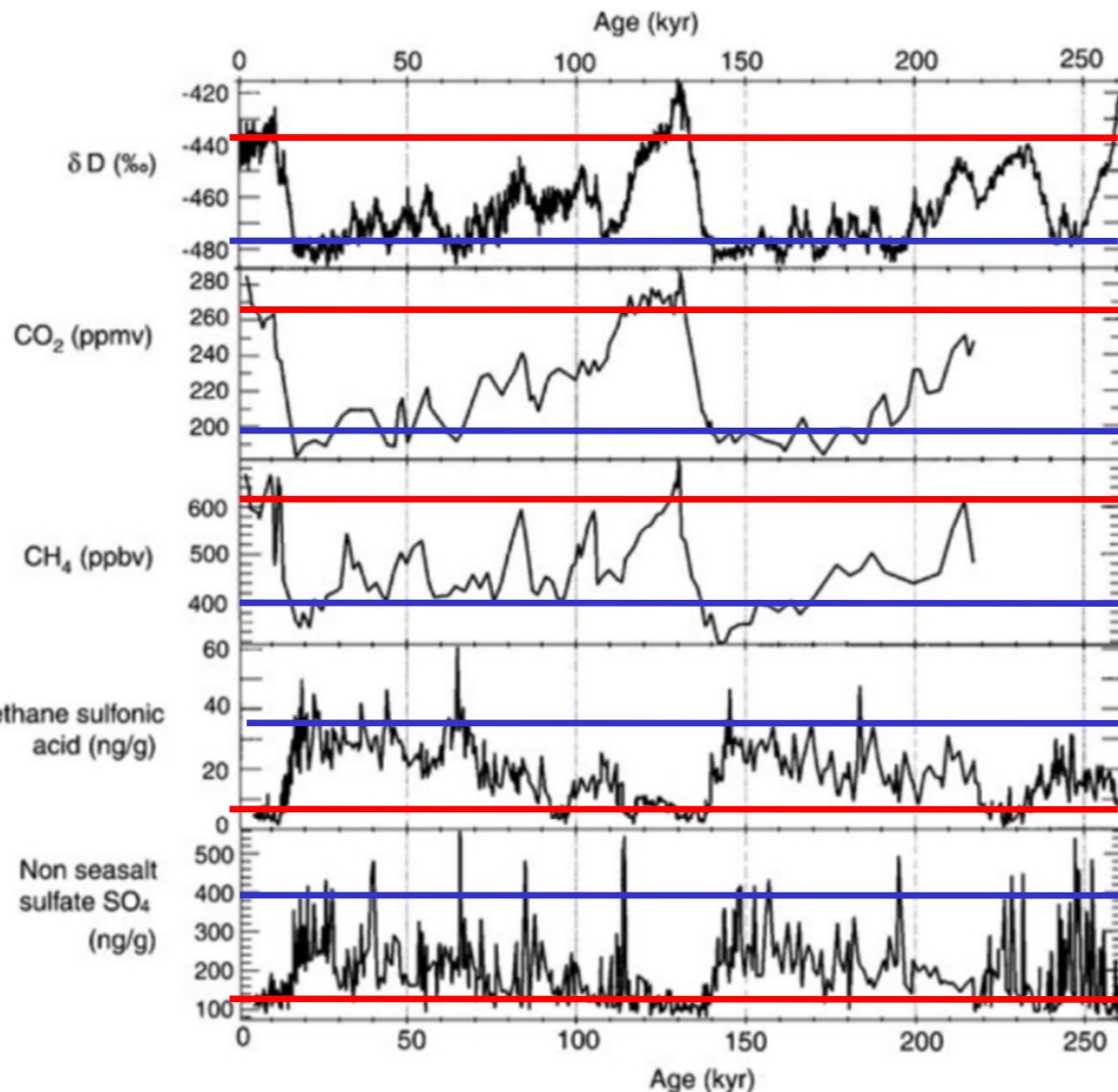
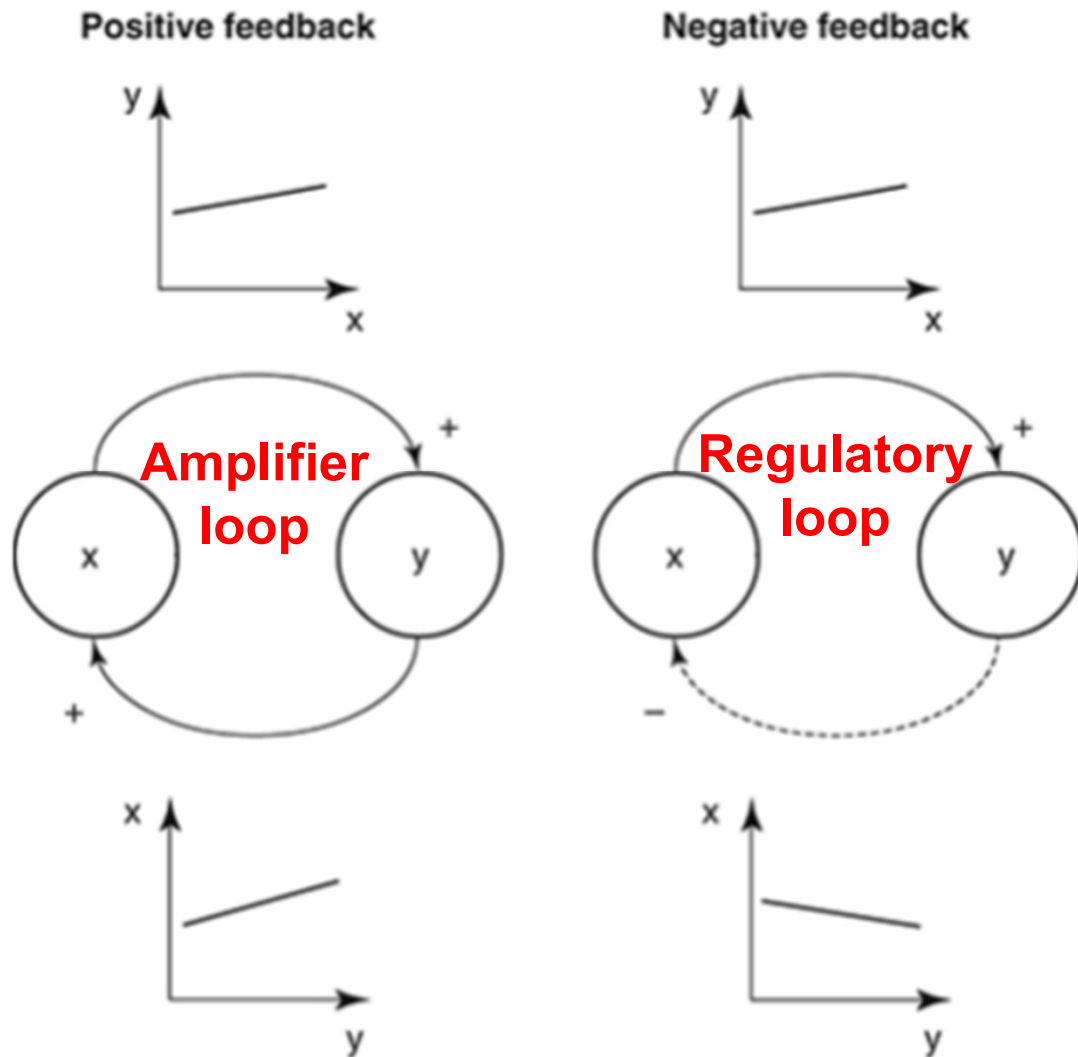


FIG. 1-2 Chemical data from the Vostok ice core. The graph of δD can be taken as a proxy for temperature changes, as described in [Chapter 18](#). CO_2 and CH_4 are greenhouse gases and vary in the same direction as temperature. Non-seasalt sulfate and methane sulfonic acid are both sulfur species existing in the particle phase, and are positively correlated with each other, but negatively with T. Major variations for all of these variables seem to correlate either positively or negatively with each other, indicating a coupled system. δD , non-seasalt sulfate, and methane sulfonic acid data kindly provided by Dr Eric Saltzman. CO_2 data are from [Barnola et al. \(1987\)](#) and [Jouzel et al. \(1993\)](#). CH_4 data are from [Chappellaz et al. \(1990\)](#) and [Jouzel et al. \(1993\)](#). (ppmv = parts per million by volume; ppbv = parts per billion by volume)

Most tracers correlate positively or negatively this is evidence for a coupled system.

There seem to be two relatively stable set points i.e. preferred climatic & biogeochemical states. The changeover between the two once triggered is rapid.



Coupled cyclic systems contain feedback cycles

2. Positive and negative feedback. A plus sign on a solid arrow indicates a direct relationship (e.g. increasing x increases y). A minus sign on a dashed arrow indicates an inverse relationship (e.g. increasing y decreases x). An even number (including zero) of inverse relationships in a loop gives a positive feedback, and an odd number gives a negative feedback.

The nicest explanation of this is in: Earth system science a very short introduction

Negative feedback cycles in technology or our body

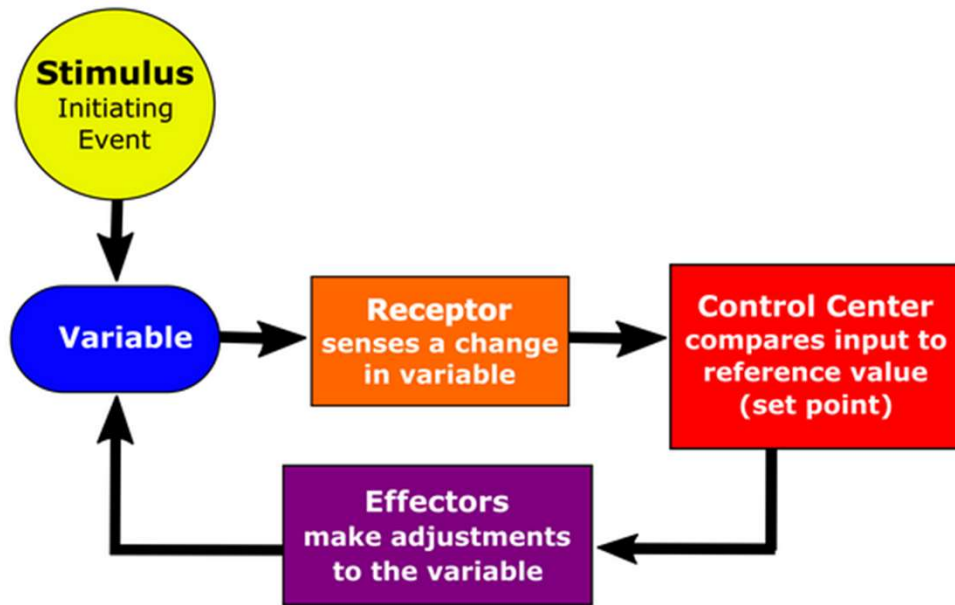


Figure 2.4. Operation of feedback loops [Credit: Open Learning Initiative, Carnegie Mellon University]

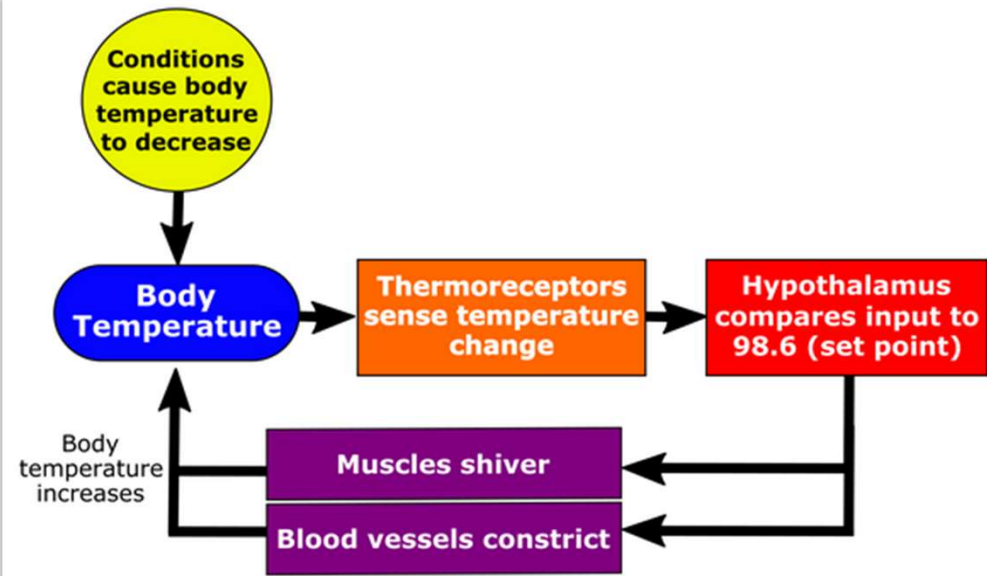
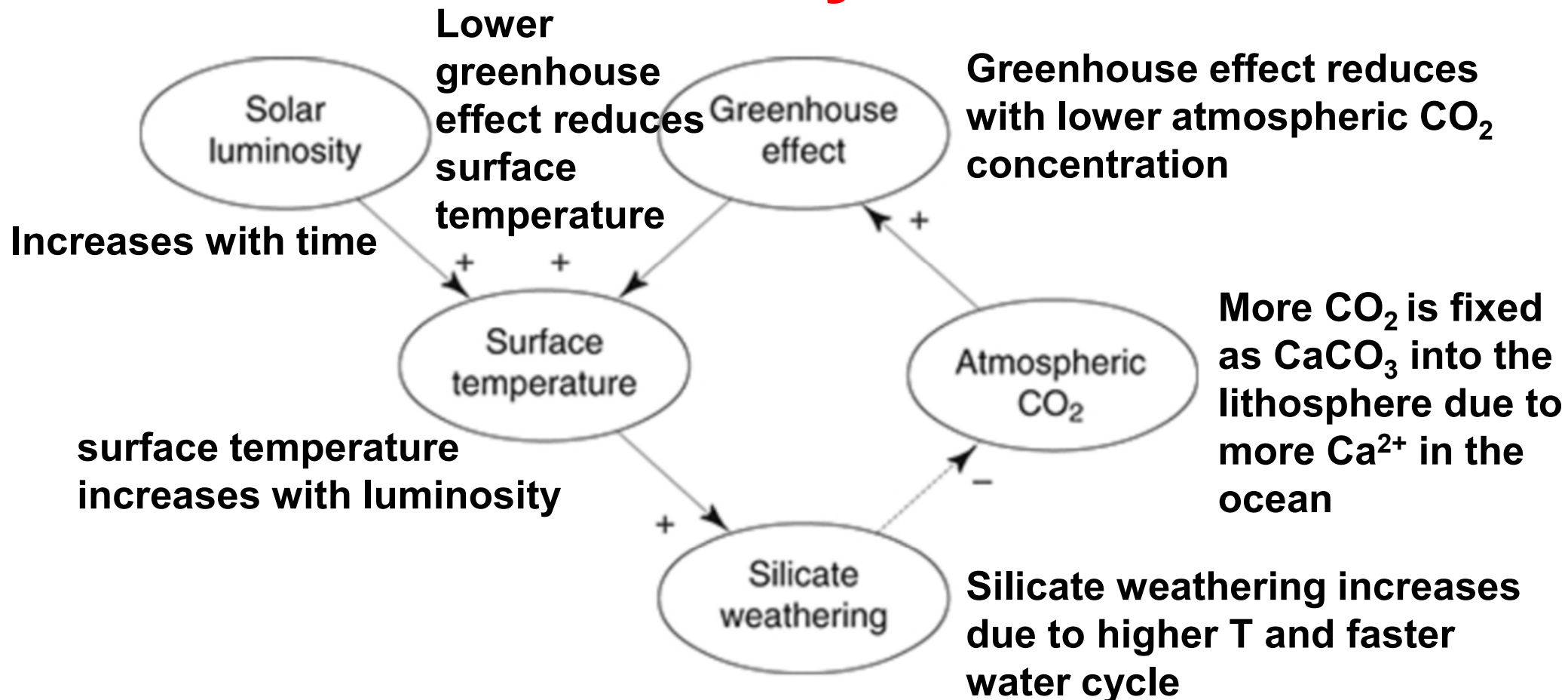


Figure 2.9. Response to Low Body Temperature [Credit: after original by Open Learning Initiative, Carnegie Mellon University, CC-BY-NC-SA]

In a negative feedback cycle a stimulus that perturbs a variable triggers a system response in which an effector restores the variable to the original conditions. Negative feedback cycles regulate and stabilize a system

The nicest explanation of this is in: Earth system science a very short introduction

Negative feedback cycles in the Earth system

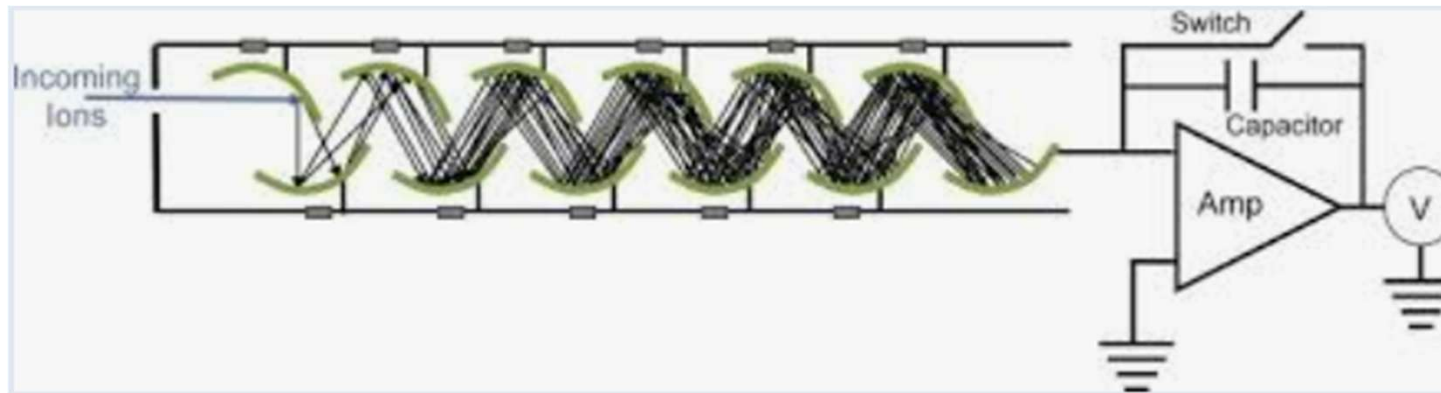


3. The silicate weathering negative feedback, with changing solar luminosity shown as an external forcing.

The nicest explanation of this is in: Earth system science a very short introduction

Positive feedback cycles in technology

The electron multiplier, photo multiplier, amplifier ...

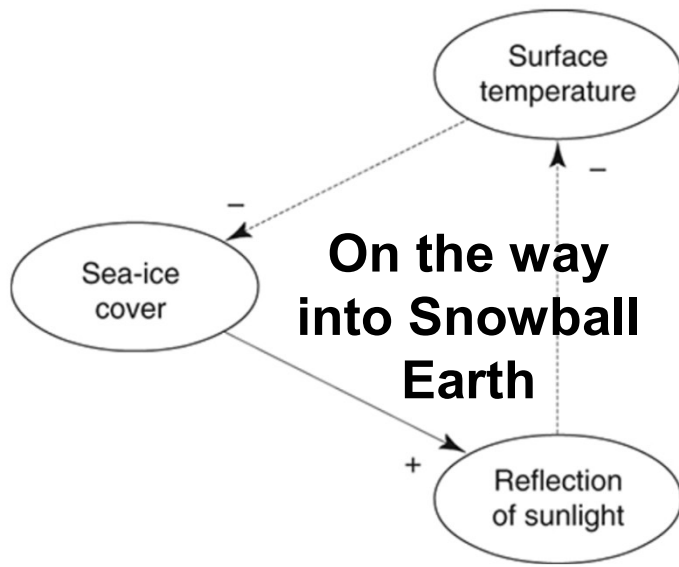


An electron or ion strikes the first diode. This leads to the emission of two electrons that strike the second diode, which emits four electrons till at the end we have more than 10^{19} electrons/second and a measurable current

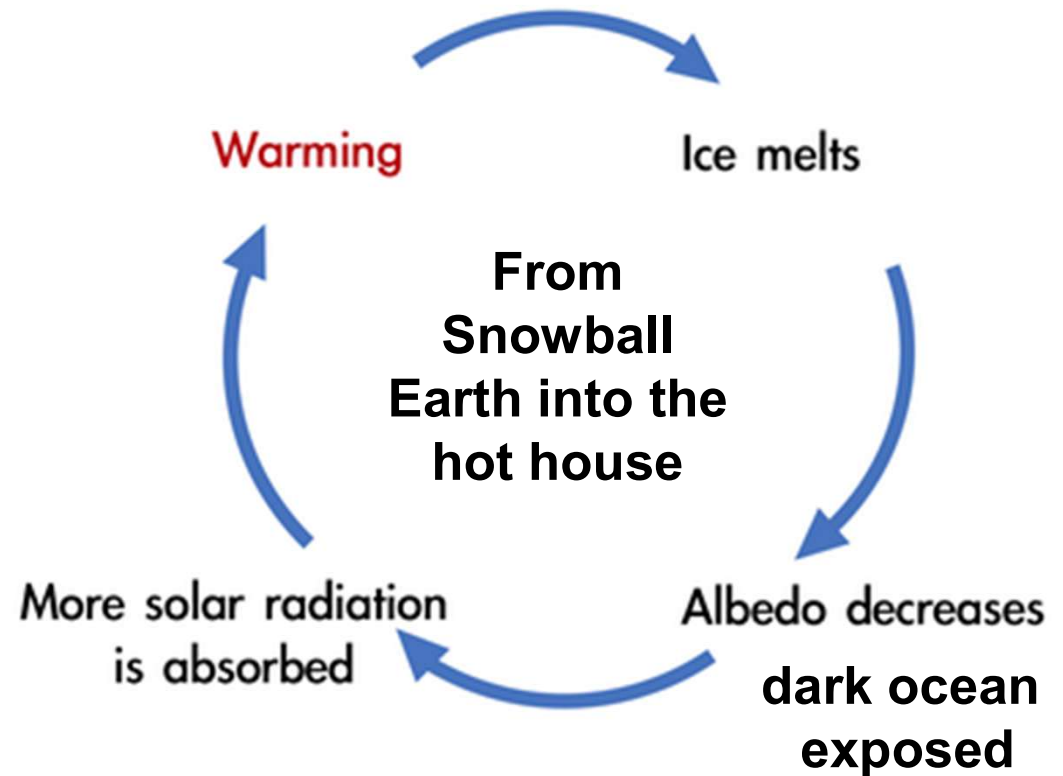
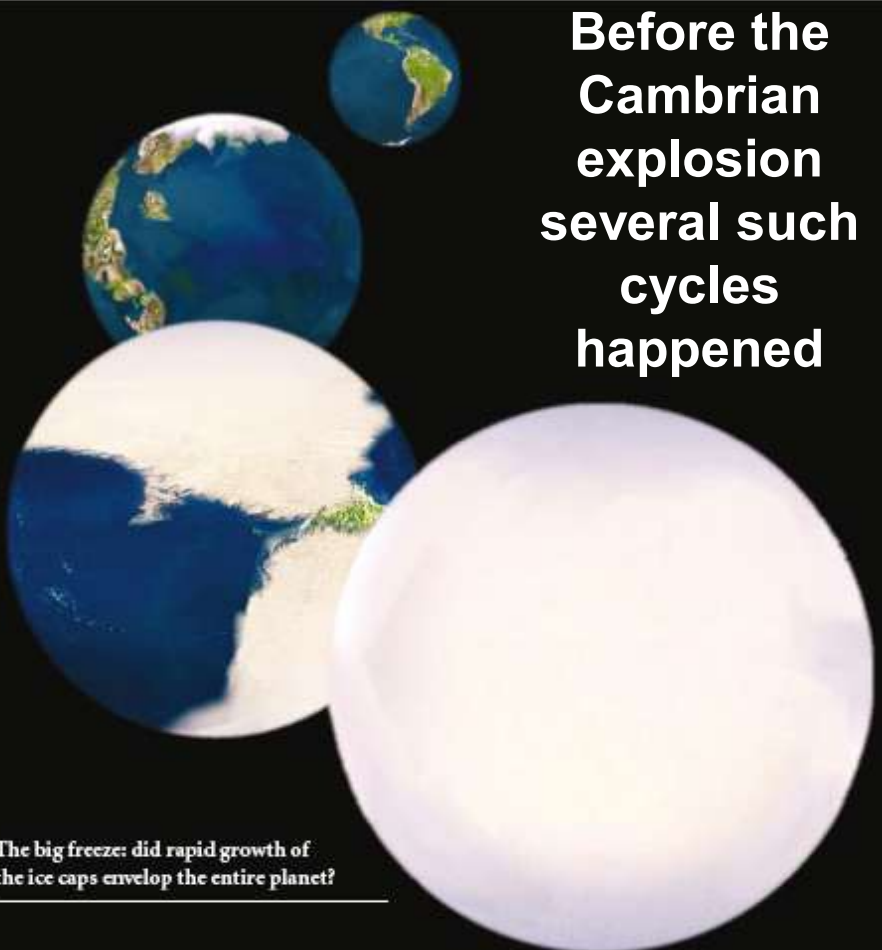
In a positive feedback cycle a stimulus that perturbs a variable triggers a system response in which an effector amplifies the original stimulus to drive the system further in the same direction. Positive feedback cycles are amplifier loops and can destabilize a the earth system.

The nicest explanation of this is in: Earth system science a very short introduction

Positive feedback cycles in the Earth system: Sea ice albedo



4. The ice-albedo positive feedback.



The nicest explanation of this is in: Earth system science a very short introduction

PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 11th August if you want to get them corrected before the Midsem exam

MCQ style

1. Which of the following is true of biogeochemical cycles?
 - a) They increase the total amount of matter over time.
 - b) They can be affected by human activity.
 - c) They create new atoms to replace lost ones.
 - d) They ensure that nutrients are only used once by organisms.
2. Which of the following best describes carbon flux density?
 - a) The total amount of carbon in a reservoir
 - b) The rate at which carbon moves into or out of a reservoir
 - c) The chemical composition of carbon compounds in the reservoir
 - d) The rate at which new carbon compounds are formed in a reservoir
3. What are the primary units used to express flux density?
 - a) g m^{-2}
 - b) $\text{g m}^{-3} \text{s}^{-1}$
 - c) g s^{-1}
 - d) $\text{g m}^{-2} \text{s}^{-1}$

MCQ style

- 1. What is meant by the "turnover time" of a reservoir?**
 - a) The total amount of a substance in a reservoir.
 - b) The ratio of the reservoir content to the reservoir sinks
 - c) The rate at which the substance moves into or out of a reservoir
 - d) The fraction of a substance that leaves the reservoir each year
- 2. Shorter residence time for carbon in a reservoir means:**
 - a) A substance is stored for a longer period in the reservoir
 - b) The substance is exchanged rapidly with other reservoirs.
 - c) The reservoir is a more significant source of the substance
 - d) The reservoir is a more significant sink of the substance
- 3. The budget equation for a given reservoir can be expressed as:**
 - a) $\text{Inflow} = \text{Outflow} + \text{Change in storage}$
 - b) $\text{Outflow} = \text{Inflow} + \text{Change in storage}$
 - c) $\text{Inflow} + \text{Outflow} = \text{Change in storage}$
 - d) $\text{Inflow} = \text{Outflow} - \text{Change in storage}$

Numerical/ Long answer style

1. In global terms 560×10^{12} kg is a relative small reservoir of carbon and 60×10^{12} kg y^{-1} is a relatively large flux. Would you expect large fluxes through a small reservoir to usually result in a long or short turnover time?
2. Calculate the turnover time and turnover rate of a reservoir containing 560×10^{12} kg carbon with a sink flux of 60×10^{12} kg y^{-1} .
3. If the turnover rate of a reservoir is 4% then what is the average turnover time / residence time of molecules in the reservoir
4. Give the general definition of a “positive feedback cycle” and a “negative feedback cycle” and explain how both types of cycles impact the climate system
5. Explain how the sea ice positive feedback cycle accelerates climate change
6. Explain how the accelerated silicate weathering climate feedback works